

Memory for Sponsorship Relationships: A Critical Juncture in Thinking

T. Bettina Cornwell

University of Oregon

Michael S. Humphreys

University of Queensland

ABSTRACT

Marketing communications that utilize partnerships between a brand or corporation and a sport, art, or community activity are thoroughly integrated into our understanding of business behavior. Despite the ubiquity of sponsorship relationships as the bases of communication platforms, we still do not understand fully how they work when successful, and how they do not work when they fail. Memory is important to the communication function of all sponsorships and a strategic objective of many. Thus, we explore memory for sponsorship relationships. The aim is to codify our progress to date in measuring memory-related sponsorship outcomes, to identify where shortcomings in our understanding remain and to move toward more complete and explanatory models of sponsorship effects. Topics discussed include: memory as a measure of sponsorship success, strong and weak memory objectives, implicit and explicit memory, interference from competitors, strategic encoding processes, cueing memory retrieval, redefining memories, and enhancing memory for sponsorship relationships. © 2013 Wiley Periodicals, Inc.

When one thinks of world cup soccer and Fédération Internationale de Football Association (FIFA), one cannot also help but think of adidas. The relationship between the international governing body of football and the sport equipment manufacturer is decades old. While the value of the relationship is difficult to quantify, one can measure memory for the relationship between the two. In fact, arguably, when one seeks to measure sponsorship outcomes, recall and recognition are natural top candidates because they are measurable and supply information that is compatible with some models of marketing communication success. However, the link between enhanced explicit memory and desirable marketing outcomes may not always be straightforward. For example, recent research suggests that sponsorship can influence one's evoked set of brands in decision making without being explicitly remembered (Herrmann, Walliser, & Kacha, 2011). Research has shown that brief exposures to limited brand information—just a logo in a context or short tag line—as typical in sponsorship can revive more established brand associations (Pham & Vanhuele, 1997). On the other hand, research has shown that direct assessment of brand attitude and purchase intention following exposure to sponsorship activities may yield little change from pretest attitudes and intentions (McDaniel & Heald, 2000). This finding suggests that even if a relationship is remembered, that this memory may not influ-

ence attitude or planned behavior. To date, examinations of subtle influences and the influence of generalized memories (Humphreys et al., 2010a) have been limited within the sponsorship context. From other areas of marketing and psychology, we can however make the case that unbidden memories may influence behavior.

Academic research on sponsorship has escalated in the last decade seemingly matching investment and utilization of the marketing platform. In the last decade more than 300 academic papers have been written on sponsorship measurement, management, policy, and concepts (Cornwell & Kwon, 2013, Working Paper). Progress has been made on many fronts but at the same time, many papers are redundant. For example, field studies where attendees are interviewed at an event venue have a disproportionately large representation in the literature. In particular, there is still a good deal that needs to be learned about how sponsorship relationships are remembered away from event contexts and in purchase and use contexts, but new thinking is in order.

Less well documented, or at the least, less publicly available, are practitioner studies—the few that are found often utilize the same thinking and measurement found in academic studies. From the practitioner side, it seems that the majority of sponsorship investments are unmeasured. It has been suggested

that this “measurement deficit” makes it impossible to effectively manage this area since proper performance results are not available (Meenaghan, 2011). This limited research repertoire described will hopefully be expanded with the stocktaking on theory offered here and with a look at what has been left unexamined. Thus, the purpose of this paper is to explore, why we measure, what we measure, and the way in which we measure memory for sponsorship relationships. The aim is to codify our progress to date in measuring memory-related outcomes and to identify where shortcomings in our understanding remain.

There are many ways in which an overview of memory research related to sponsorship might be organized. Following the basics of memory, one might discuss the structure of memory—including short- and long-term memory, processing and control of memory. Alternatively, one might look at memory research within the various areas of sponsorship and attempt to categorize the type of memory work being done. Our decision was not to take these paths, but to focus on areas that are most informed by research that addresses memory processes, as well as areas with the highest relevance not only to sponsorship but also to the more general area of indirect marketing communications (Cornwell, 2008). Therefore, much of the thinking here could easily apply to low message media landscapes including brand placement, ambient marketing, interactive marketing, and social media. The challenge of measurement, learning, and enhancement/retention are the foci, and the topics covered include:

1. The Importance of Memory as a Measure of Sponsorship Success
2. Memory Objectives, Strong and Weak
3. Implicit and Explicit Memory
4. Interference: What Happens when Competitor Information is Introduced?
5. Strategic Encoding Processes
6. Cueing in Memory for Sponsorship
7. Memory Experiences Combining With, Defining, or Redefining Other Memory Experiences
8. Retaining and Enhancing Memories

Irrespective of our approach to organization, given the breadth of topics and potential application of new theories and thinking from memory to marketing, some things will have to be left from the discussion.

THE IMPORTANCE OF MEMORY AS A MEASURE OF SPONSORSHIP SUCCESS

Memory outcomes for sponsors are associated with sponsorship success. Typically these are measured as improved memory for the sponsor brand, and often memory for the sponsor–sponsee relationship on the part of viewers or attendees of the music, art, or sport event. These outcomes also correspond nicely with many models of marketing communication suc-

cess (e.g., Keller, 1993, 2003). Memory for a sponsorship relationship, that is, for the link between the event and the sponsor, correlates less well with traditional advertising models. Advertising in traditional media is typically concerned with the image, attitudes, or purchase intent resulting from the advertisement being presented in broadcast programming or within a contextualized environment in magazines, newspapers, or online. In advertising research, there is little, if any concern about remembering a relationship between the brand and the program or media vehicle *per se*. A high hurdle in sponsorship, to explicitly remember the link between the event property and the sponsoring brand, is sometimes a worthy goal. For example, if the brand seeks to develop associations to a cause or charity in order to build their corporate social responsibility record in the mind of consumers, explicit memory of the sponsor–event relationship may be key. In some other areas of sponsorship, this demanding level of memory may be more than is needed to further marketing-related goals. For example, in-store displays tied to the sponsorship may cue behavior regardless of whether or not they also produce recall or recognition of the sponsor–event exposure. We know much less about these memories that may be cued by an in-store context, packaging, a spokesperson, or even something as subtle as color or design.

MEMORY OBJECTIVES, STRONG AND WEAK

Applied work in memory is naturally oriented to communication objectives. Learning about a brand and a brand’s characteristics is typically expressed in the sponsorship literature as building brand awareness. The vast majority of sponsorship measurement to date in academic publications could be classified as capturing strong learning objectives via cued recall and recognition, and to a lesser extent, free recall (Cornwell, Humphreys, Quinn, & McAlister, 2012). As an example, Johar and Pham (1999) utilized press release announcements to expose participants to sponsorship relationships. Following exposure to four press releases, each participant completed a rating task and a filler task. Participants were cued with the event for the sponsor and were also asked about their confidence in this response. In this task, 32% reported feeling sure that they recalled the sponsor and 56% of these had actually identified the true sponsor.

The Johar and Pham (1999) experimental procedure is related to popular field procedures where someone who has recently watched or attended an event is asked about the sponsor(s) of that event. In both situations it is likely that the context where the person learned about the sponsor–event relationship will be used as part of the retrieval cue and this use of the contextual information should make the recovery of the relationship easier. Context likely influences most retrieval in field studies where spectators and event

attendees are pre- or postevent study participants, however, unless we learn of their past exposures, we know little of how they make use of the current or a prior context in recall or recognition.

An alternative to context-dependent retrieval is to arrange conditions so that the relevance of the recent exposures to the question being asked is not obvious (Humphreys, Bain, & Pike, 1989). As discussed subsequently, this type of testing might be considered an examination of implicit memory. In lab studies this is achieved by directing attention away from the recent exposure context. For example, after stimuli exposures such as press releases containing sponsorship information, questions that can be answered using the information from the press releases are embedded in a series of questions that cannot be answered using information from the target stimuli (e.g., those particular press releases). The more naturalistic situation which this might correspond to is a situation where some time has elapsed since attending or watching an event, thus making it difficult for the person being questioned to reinstate the context where the link was learned (see Humphreys, Tangen, Cornwell, Quinn, & Murray, 2010b for examples).

Thinking about the important role of context has a long tradition in the area of eyewitness testimony. Early work in this area showed that context reinstatement significantly increases accurate identification of a person in question and influences one's confidence in this identification (Krafka & Penrod, 1985). This stream of research continues to show a strong supporting role in memory of context as a cue (Wong & Read, 2011). This suggests that research having a strong context cue will likely show stronger recall than those studies designed to avoid context effects. This facilitated recall is, however, typically against the strong memory objective of explicit recall or recognition.

Even if the memory link between the sponsor and the event is not strong enough to pass the high hurdle of explicit recall, it is possible that this link or other links that were formed while watching or attending the event might still influence behavior. For example, after attending a NASCAR event, the demonstration of a tire performance during the race might make a particular brand come to mind or seem appealing in a retail context. While a person might not have a direct recollection of the event or even the performance demonstration, there is facilitation of a brand-related task. For example, Herrmann, Walliser and Kacha (2011) examined memory effects of tennis spectators and found that even when sponsoring brands were not recognized, they still were present more often in a consumer's consideration set. The mere exposure effect (Zajonc, 1968) is another example where facilitation of sponsor goals may occur under limited explicit memory (see Cornwell, Weeks, & Roy, 2005 for a discussion). In this case repetition tends to build positive affect that we measure as liking. Reber, Winkielman, and Schwarz (1998) find that a two-step attributional theory of processing fluency, where exposure to a stimulus leads to greater fluency

on a second exposure, explains this report of liking. The idea is that enhanced fluency may not be correctly attributed to the prior exposure but rather misattributed to features of the stimulus (e.g., look or style). This facilitation of some action or behavior without conscious recollection is the essence of implicit memory (Schacter, 1987). This facilitation, or what we call "weak" memory objectives, has not been the focus of the majority of sponsorship research.

IMPLICIT AND EXPLICIT MEMORY

Testing

As will be reiterated in several ways, the way one tests for memory of a sponsorship relationship is key since it can reveal or hide relevant strategic information. The distinction between an implicit and an explicit memory test is clear. From the psychology perspective, instructions for the implicit test make no reference to the learning episode (the event or activity is not mentioned) whereas the instructions for the explicit test make reference (Do you remember seeing a logo for a brand of sunscreen at the volleyball game?). A definition of implicit and explicit memory in psychology stems from the nature of the test employed but defining the nature of these memories is less concise and agreed. Schacter (1987) explains explicit memory as "conscious recollection of recently presented information, as expressed on traditional tests of free recall, cued recall and recognition" (p. 501). In contrast, he describes implicit memory as the facilitation of some task that is "attributable to information acquired during a previous study episode" (Schacter, 1987, p. 501). Unlike in the marketing literature, there is reluctance to call this unconscious or unaware memory. That is, simply asking about the sponsor of an event under conditions where no information is supplied about the learning context, especially where the participants are unlikely to supply themselves with this information, can be considered as an implicit test. Nevertheless, this is a question which has a correct answer (we will refer to it as a definitive question) and attempts to answer it may evoke somewhat different processes (e.g., a postretrieval evaluation of the answer) than answering a question which does not necessarily have a correct answer such as asking people to generate brand names that would make good sponsors for the event. Importantly, the nature of the memory challenge put to study participants is very strongly related to the way in which they attempt to find information.

Implicit and Explicit Memory Retrieval Attempts

Although one can be clear in establishing a test of memory that is either explicit (referring to a particular episode) or implicit (with no specific referent), it is

more challenging to learn the individual's pathway to retrieval of information. Study participants can retrieve in accordance with the implicit instructions (implicit memory retrieval) but become aware that what they are retrieving occurred in a specific episode. Therefore, awareness or seeming lack of awareness, does not necessarily distinguish between implicit and explicit memory. It is also possible that people can use information about a specific episode in attempting to retrieve information without being aware that they are using that information. Thus, if the memory test takes place a few minutes after the learning episode in substantially the same surroundings and conditions (e.g., in the sports arena at halftime) one cannot rule out the unaware use of contextual information in accessing memory. Finally, in an attempt to answer an implicit test question people may spontaneously reinstate a context in which they may have previously encountered the sought-after information.

One way available to sponsorship researchers to examine the difference in retrieval is referred to as retrieval intentionality. If there is a difference between implicit and explicit retrieval instructions this supports a difference between the instructions and possibly the type of information accessed. Here everything is held constant but the type of test. That is, the cue and the response are the same, but the test instructions differ. For example, compare cued recall instructions using word stem cues with instructions to produce the first word that comes to mind that completes the word stem. Stem completion test were brought into advertising some time ago (Duke & Carlson, 1994) but are not frequently seen in sponsorship research.

In typical sponsorship research if we ask a question about the sponsor–event link as a definitive question it may or may not make reference to a specific episode. If more is to be understood about retrieval attempts then more lab work in sponsorship is implied. To vary test instructions, control of the testing environment is needed. The gold standard would then be movement between field and lab, and though many may advocate the lab then field sequence, some of the communication processes in sponsorship may need to be first established as occurring before being dissected in more academic analysis.

In order to achieve the strong memory objectives of sponsorship, more research is required on the ability to answer definitive questions when the use of contextual information is unlikely. In addition, in order to achieve some of the weak objectives in sponsorship, more research is needed on when or how the sponsorship experience can change answers to implicit questions—in the absence of being able to answer an explicit question. Hopefully, this discussion of the relevant conceptual issues involved in discriminating between definitive and indirect questions and between explicit and implicit questions will help in this endeavor.

INTERFERENCE: WHAT HAPPENS WHEN COMPETITOR INFORMATION IS INTRODUCED?

Lacking in many examinations of sponsorship and related indirect marketing platforms is a thorough understanding of interference. Because sponsorship is embedded and integrated in the social milieu, any sponsorship episode has high connectivity with the associates of the event or activity. Over time, various exposures to news, products, and experiences establish additional associations. On the one hand, this might provide many paths for successful memory of the sponsor brand and related characteristics and qualities; on the other hand, it invites various connections, which may lead to confusion, especially with a direct competitor having similar characteristics.

Take for example; the instance of sponsorship-linked advertising where a sponsor of a sporting event leverages their investment by placing thematically tied advertisements in media both during the event and following it. This leveraging approach is not limited to a true sponsor. If a direct competitor places an ad during the same time window, there is the chance that the two brands in the same category could be confused. Moreover, if this nonsponsor, competitor brand also utilizes a sport theme; the chances of interference may increase (Kelly, Cornwell, Coote, & McAlister, 2012). Following are additional areas where studies examining aspects of interference have been conducted. As well, areas needing future longitudinal investigation are discussed.

Ambushing

One of the areas of sponsorship memory where we know less than is needed is in the area of the effects of ambushing. Ambush marketing, which has been defined as the formation of an association with an event without securing official sponsorship rights (Sandler & Shani, 1989), has evolved over the past decade. Original concern was with the intent of the ambusher to gain association with the event and to be mistaken as the “true” sponsor. The goal of misattribution as “official sponsor” has gradually been supplanted by the fact that many ambushers may simply seek awareness via association with a sponsored event and to capitalize on goodwill generated by the event or property (Burton & Chadwick, 2009). This evolved view comes full circle to early arguments by Meenaghan (1996) that ambush marketing could be viewed (from the perspective of the ambusher) as legitimate competitive marketing practice. It also encompasses the fact that memory measures may capture “incidental ambushers” (Quester, 1997) that may have no intention of ambushing but are a naturally supplied probable answer to the question “who sponsored this event?” Nonetheless, ambushing activities including use of phrases and images associated with the event or activity, purchase of advertising time within the event broadcast, presence in and

around the venue, as well as use of consumer promotions and congratulatory messages (McKelvey & Grady, 2008) may hold memory implications for the true sponsor.

One memory-related ambushing question, long left unexamined, was what to do once ambushed. Advice to “name and shame” the ambusher relied on studies showing that negative attitudes may be associated with the behavior (Dalakas, Madrigal, & Burton, 2008). In contrast, Humphreys et al. (2010c) found that when sponsors utilized counterambushing messages in response to ambushing attempts, the result was to build an association between the ambusher and the event; which at times resulted in the ambusher being identified as the sponsor. In support of practice and in support of finding a balance between corrective action and detrimental attention to ambushers, future research needs to address both memory and related attitudes toward ambushing simultaneously.

This brings forward, a remaining question regarding the influence of ambushing and that is to clearly understand the gain in memory for the ambusher in relationship to any detrimental effects for the true sponsor. That is to say, an individual may at the time of the test of memory, select an ambusher foil from a list of possible sponsors but still hold a memory for the relationship between the event and the true sponsor that may result in, for example, positive, in-store behaviors. In general interference is not destructive. That is to say, interference by communications from direct brand competitors or even noncompetitors with a link to the event may disrupt access to information about the true sponsor but it does not substantially destroy or replace that information. Memories and the access to memories must both be considered. If the memory is still there, then access to it may be possible.

The best evidence against the nondestructive nature of interference in recall comes from research using what is called the AB AD paradigm. In this exposure paradigm, a word, “A” (or perhaps a brand name) is paired with another word, “B” (or perhaps an event or an adjective) and embedded within a list containing similar pairings (the AB list). Then in a subsequent list the initial word, “A” is paired again with another, “D,” (the AD list). Associative interference occurs when the recall of B and/or D to the A cue is reduced relative to the situation where only a single AB or AD list has been studied. However, there is almost no associative interference (a reduction in the recall of the other pair member) when recall is cued with either the B or D term (Barnes & Underwood, 1959). Further evidence against destruction of memory by interference is found in associative recognition paradigms where pairs such as AB, EF, and GH are studied and then the participants are asked to discriminate between intact pairs—pairs they have seen before (e.g., AB) and rearranged pairs—elements seen before but in a new relationship (e.g., EH). When associative interference is introduced (e.g., studying pairs AD and CB after studying AB and testing AB as an intact pair) there is a tendency to

increase the probability of correctly identifying AB as intact relative to a situation where AB is studied by itself (Dyne, Humphreys, Bain, & Pike, 1990). The point is that memories are typically still there and as we discuss next, may be spontaneously recovered with the right cue.

Spontaneous Recovery

In a 1948 paper, Underwood reported that participants who had learned a list of pairs (each word “A” paired with word “B” was denoted as an “AB pair”) and then learn a second list where the cue stays the same though it is paired with a new target (AD) varied in their responses depending on the time delay. Participants were given a test after one minute, five minutes, 24 hours, or 48 hours. On this test, participants were asked to produce the first of the two response words that came to mind. The recall of the List Two target declined from one minute to 48 hours while the recall of the List One target increased slightly from one minute to 24 hours. Underwood (1948) referred to this finding as “spontaneous recovery.” Spontaneous recovery in the sponsorship context occurs when an event, A, is paired with a sponsor, B, and then in a change of sponsorship a new company/brand, D, becomes associated with the event, A. In an immediate test of memory, the new sponsor, D, may be recalled but over time and without substantial rehearsal or repetition (e.g., via advertising) of the new relationship between A and D, memory for the original AB pairing arises. In a review of the early work on spontaneous recovery, Brown (1976) determined that absolute recovery (where memory for the initial pairing is greater on the delayed test than on the immediate test) is relatively rare. However, relative recovery (memory for the initial pairing declines at a slower rate than memory for the new pairing) is common (for a more recent review see Wheeler, 1995).

Recent evidence for the occurrence of spontaneous recovery in the sponsorship area is found in both field and lab settings (McAlister, Kelly, Humphreys, & Cornwell, 2012). This potential for spontaneous recovery is very important in the current sponsorship market where the vast majority of sponsorship properties have been held previously. A long-standing relationship held by a well-known brand may be recovered after a new contract with a new sponsor has been made and in place for some time, even years (McAlister et al., 2012). This suggests that communications plans need to take into consideration this probability and to measure the extent of rival brand information over time. This would be particularly important when the old sponsor and the replacing sponsor are in the same category and share many characteristics.

Confusing the Competitor for the Sponsor

Sponsorship research in the popular press often questions the value of sponsorship when a brand, not having

a prominent sponsorship role is thought to be the event sponsor. Disturbing to the brand making a significant sponsorship investment is the research that shows that a foil, a brand not sponsoring and sometimes entirely unassociated with the event can be thought to be the sponsor. This seems particularly worrisome when the brand is in the same category. Is it really that bad for the true sponsor? Conditions have to be carefully examined to tell whether information about the sponsor–event link is still available even though there is also information about the competitor–event link. Information about the competitor–event link may occur because the competitor has been specifically associated with the event (see the previous discussions about ambushing and spontaneous recovery). It is also possible that what appears to be a learned link between the competitor and the event may arise from confusion caused by the similarity between the competitor and the sponsor (e.g., PowerAde and Gatorade are used in similar ways, look similar and are regularly found in similar types of contexts).

Disentangling the effects due to similarity from other effects is especially difficult using the forced-choice procedure where the event may be provided and the person must choose which one of several alternatives was the sponsor. One problem with forced-choice recognition is that what appears to be memory confusion may simply be the result of guessing. Marketing researchers tend to want to address this problem by allowing the participant to respond, “don’t know” so not to force an answer from a participant. The issue with “don’t know” responses is that they introduce another source of variability. As is established, there may be variance in willingness to use the “don’t know” response. The possibility of such variance makes it impossible to compare individuals or conditions in terms of accuracy. Also in the forced-choice task there are several determinants of errors, which are unrelated to the similarity between the target and the distracter. For example, past basic research in psychology shows that high-frequency distracters which occur frequently in the language are more likely to be chosen than low-frequency distracters which occur less frequently in the language both in single item (yes/no) recognition and in forced choice tests (Glanzer & Bowles, 1976). In brand choice, there is an apparent tendency to select distracters that have higher market prominence (Johar & Pham, 1999). This predisposition toward options based on preexperimental considerations results in a bias, or “base rate effects.”

Putting aside base rate effects of forced choice, this test still does not give a clear indication of target-distracter confusability (true sponsor and direct or indirect competitor). That is, the probability of choosing the distracter is not only going to be related to the similarity between the distracter and the target, but the choice is also going to be determined by how well the target has been learned. This degree of learning is often conceptualized as target strength or familiarity. If the target is well learned, there may be no errors, but if

the target is poorly learned the choice of the distracter will approach 50% regardless of target-distracter similarity. This is what the “don’t know” response option is thought to solve but does not. In short, with two choices and little knowledge about who the true sponsor is, the results of the test will tend toward guessing levels (50/50). Under these conditions, a foil might be seemingly more associated with the brand than may be the case.

In addition, there is a curious effect of the correlation between target strength and distracter strength. Tulving (1981) used very similar distracters and compared performance on two different kinds of forced choice tests. Surprisingly, memory performance was better in the condition where the target and the distracter were similar to each other. The standard explanation for this effect depends on the assumption that when a target is similar to its paired distracter (often denoted as target, A and distracter, A’) the strengths of the two items will be similar, that is, they will be positively correlated (Heathcote, 2006; Hintzmann, 1988). The result will be a greater ability to discriminate between target and distracter than when a strong but nonsimilar distracter is used (e.g., in word pairs, the choice is between A and B’, where B’ is similar to a studied word B but not similar to A).

These problems with the forced-choice procedure can be overcome using a version of a yes/no task. In this way, the question is set so that all is required is a yes or no response from a study participant. In this design, after studying a list of brand–event pairs, participants are tested on (1) target brand–event pairs (e.g., Buick and Golf Open), (2) brand–event pairs where the event was studied with a brand that is similar to the target brand (similar distracters—the studied pair is Lexus and Golf Open and the tested pair is Buick and Golf Open), and (3) brand–event pairs where the event was studied but the brand was not studied and is not similar to any studied brand (dissimilar distracters—the studied pair is Sony and Golf Open and the test paired is Buick and Golf Open). Counterbalancing is used so that each brand is used equally often as a target, a similar distracter and a dissimilar distracter. The probability of saying that a dissimilar distracter is intact (the pair seen before) provides a measure of base rate effects. The difference between responding that a pair is intact to a similar and a dissimilar distracter is the measure of target-distracter similarity, and the probability of responding that a pair is intact to a target is the measure of target strength. In this way, one is able to disentangle the various effects.

As we previously noted interference is primarily nondestructive. Thus, the probability of calling a target intact may be high (high-target strength) and at the same time the probability of calling a similar distracter intact may be high (high-target distracter similarity). Thus, people can perceive that there is a link between a competitor of the sponsor and the event either because the competitor has been specifically linked to that event (e.g., advertising during the time of the event) or

just because the competitor is similar enough to the sponsor for confusion to occur. This may well benefit the competitor but because it probably does not damage the link between the sponsor and the event it may not have that great an impact on memory for the true sponsor.

STRATEGIC ENCODING PROCESSES

People seem not to invest effort in marketing stimuli, so within marketing and sponsorship interest has primarily focused on low-level processing and improved memory via repetition. There has been, however, some interest in “depth of processing” and conscious processes of encoding within brand placement. Also, the concept of the fit between the sponsor and the event is closely linked to ideas about conscious strategic processing within the memory literature.

Information processing studies in marketing concerned with encoding often have roots in the depth of processing research originally conducted by Craik and Lockhart (1972). The concept of processing depth is convenient shorthand for thinking about the kind of processing strategies that enhance the learning of individual items. For example, asking study participants to rate an item for pleasantness, asking if it is larger than a breadbox, or asking them to rate how frequently the word occurs in the language can increase depth of processing but asking a question about the orthography or phonology of the word is considered to be more shallow processing and does less to enhance memory. Balasubramanian, Karrh, and Patwardhan (2006) argue that processing depth along with stimulus/execution variables and individual factors shape the influence a brand placement will have. Brand placements in broadcast programming and even video games (Schneider & Cornwell, 2005) have the potential to integrate with the story line. This same potential is being recognized and utilized in sponsorship of sports, art, and entertainment. For example, in 2012, Proctor and Gamble’s Tide brand laundry detergent drew connections to the family support that Olympic athletes experience the “P&G Family Home” where among other things, the laundry could be done. Depth of processing is not, however, as helpful in thinking about the learning of links between two or more items as are the concepts of item and relational information (to be discussed shortly).

As mentioned, depth of processing and repetition with some boundaries are generally thought to increase memory quality, however, as discussed by Balasubramanian, Karrh, and Patwardhan (2006) a great deal also depends on the nature of the stimuli. In brand placement, the modality of placement (visual, verbal, or both) is of concern, and is similarly of interest in sponsorship but has not been the focus of researcher attention. Another stimulus/execution characteristic of particular importance for the loosely structured sponsorship context is the role that provided information

plays relative to existing knowledge. Einstein and Hunt (1980; Hunt & Einstein, 1981) argue that distinctiveness encoding applies to the processing of item information, while similarity encoding applies to the processing of relational information. Importantly, although item information and relational information support each other, each can be an independent contributor to recall. For example, a person attending tennis events regularly may have acquired a great deal of relational information on the category of tennis events. The common characteristics of tennis events: nets, stands, announcers, and sponsors, may support relationships between individual items that stem in part from the shared context. For a person to remember a car company sponsor and more importantly a particular sponsoring tennis equipment manufacturer, it would be helpful if they had an experience that communicated item-specific information—something that enhances distinctiveness of the sponsor.

Relational information activates the general class or category to which a specific stimulus belongs. Item information is then used to search within this limited group of representations. Both relational and item information contribute to memory retrieval processes, but in different ways and may be useful in deciding how best to communicate sponsorship information (Weeks, Cornwell, & Humphreys, 2006). More recently Hunt (2013) has argued that relational and item-specific processing can be framed as the processing of difference in the context of similarity and that this kind of processing can produce very good memories. At first glance this might sound simply like the age-old argument in marketing for brand differentiation and past research shows sponsorship supports differentiation (Cornwell, Roy, & Steinard, 2001). More specifically in this theoretical argument, memory is strongly supported by difference within a context of similarity. For example, banks are commonplace sponsors across many professional sports and their offerings and relationships to sport are similar, as is their use of a corporate logo—most communicate strength and history—but Aflac, in a sea of corporate stodginess, communicates with a duck. While there may be good reasons to choose or not to choose a duck (as opposed to a rock, a key, a bull, or a stagecoach) as a visual associate for a bank, the point here is that it will likely be remembered as different in this context of similarity.

If a brand has abundant item-specific information in a context, an objective might be to better establish how they relate to the context. In contrast, if the brand is tightly woven to the sponsorship context, they may want to emphasize some unique characteristic so to stand out from the crowd. Thus, the communication approach to support memory depends on the relationship between the sponsor and the context. Importantly, if a brand has various sponsorship properties in its portfolio, a single-communication approach may not serve well in all contexts. A refined memory measurement strategy would take into account the nature of the supplied information relative to the context and

expectations of already existing item and relational information. Moreover, while this is related to marketing objectives of brand differentiation, it is distinctly about a communication needs relative to the context at hand.

The congruence/match or fit between the sponsor and the event is widely regarded as improving the memory for the sponsor–event link and improving sponsorship outcomes (e.g., Becker-Olsen & Hill, 2006; Becker-Olsen & Simmons, 2002; Olson & Thjomøe, 2011; Pracejus & Olson, 2004; Rifon, Choi, Trimble, & Li, 2004; Weeks, Cornwell, & Drennan, 2008). It is extremely difficult to tell whether this is an encoding effect or a retrieval effect. That is, congruence could be instrumental in storing a better memory or, alternatively, if someone is asked about the sponsor of a track meet, the preexisting connection between track meet and running shoes might help to trigger the memory for a sponsor who markets running shoes. One study, however, clearly shows an encoding effect. Cornwell, Humphreys, Maguire, Weeks and Tellegen (2006) showed that when the sponsor and the event were unrelated (did not match/fit) the articulation of a very general and nonspecific link between the sponsor and the event improved memory for the sponsor–event relationship.

Repetition

As the psychology literature established (Humphreys et al., 2010a; Nelson, 1977) and marketing literature has demonstrated (e.g., Appleton-Knapp, Bjork, & Wickens, 2005; Janiszewski, Noel, & Sawyer, 2003; Schumann, Petty, & Clemons, 1990), spaced repetition even at low levels of processing and/or low levels of involvement establishes good memories (e.g., Hawkins & Hoch, 1992; Hawkins, Hoch, & Meyers-Levy, 2001; Schumann, Petty, & Clemons, 1990). In sponsorship, message repetition has been shown to influence perceptions of congruence between the sponsor and event (Dardis, 2009). Furthermore, in advertising the cumulative value of repeated exposures is known to be related to the spacing between exposures (Janiszewski, Noel, & Sawyer, 2003). The usefulness of repetition in advertising has also been documented as being thwarted in part when presentation of similar products results in interference (Burke & Srull, 1988). Sponsorship-linked communications, like advertising, present brand and corporate information over time and with various types of exposures (e.g., logo in a stadium, mention of sponsor by spokesperson). Sponsorship may follow some of the established expectations but may also differ due to the way it is embedded in events and programming.

Aspects of repeated exposure raise important public policy questions regarding the ways in which information is accumulated and later utilized in decision making. Many products facing control or the threat of control in terms of advertising regulation turn to spon-

sorship to maintain broadcast communications. This has been the case for tobacco for many years (Cornwell, 1997) and has become an issue of discussion for alcohol, gambling, and now “junk foods” such as fast food, snack foods and sweet, nonalcoholic beverages (Maclean & Bonington, 2008). Research suggests that adult products such as tobacco, alcohol, and gaming (Cornwell, 1997; Jones, Phillipson, & Barrie, 2009; Lamont, Hing, & Gainsbury, 2011) reach youth audiences and associate these products with images of active, fun-filled lifestyles which may be associated in the mind of the observer with health. The same can be said of food and drink high in sugar, salt, and fat, and these products for the most part may be purchased and consumed by children.

Research on repeated exposure via sponsorship for products in these sensitive product categories has mainly considered explicit memory or awareness (e.g., Jones, Phillipson, & Barrie, 2009), attitudes and purchase intention (e.g., McDaniel & Heald, 2000), perceptions of appropriateness of the sponsor for sport (e.g., Danylchuk & MacIntosh, 2009), and the potential reinforcing association of tobacco and alcohol pairing (Jiang & Ling, 2011). While these perceptions, intentions, and attitudes are of concern, so to are repeated exposures that result in changes in judgments or behavior that may be unbidden and unexamined (Humphreys et al., 2010b). Because sponsorship communications are embedded, and become part of the experience of attending or viewing an event, they are less likely to invoke a person’s persuasion knowledge defenses (Friestad & Wright, 1994). For example, as a product such as fast food is paired with event images, such as those in sport suggesting health and fitness, no explicit change in attitude or awareness may arise but as implicit memory testing suggests, some change to subsequent judgment or behavior may result.

CUEING IN MEMORY FOR SPONSORSHIP

Before examining the issue of cueing in sponsorship memory research, it is perhaps useful to draw emphasis to the importance of cues in memory. We have a working sense of the value of cues to memory, for example, having a hook near a door where the keys always go can help one remember both to put them there when not in use and to look there when exiting the house. Cues are important in memory since they initiate search of memory, they support recall and recognition, and they influence the structure and nature of retrieval. Cues are so important that very memorable events can be temporarily forgotten without the proper cue to support retrieval. As an example, Smith and Moynan (2008) demonstrate that an unusual and thus highly memorable experience (being shown a list of expletives during the course of a memory experiment) could be made inaccessible by exposure to interfering tasks after exposure to the words. On the other hand, when Smith and Moynan (2008) provided an appropriate cue it supported

recovery of the list words. This reflects on early thinking in psychology (Tulving & Pearlstone, 1966) regarding the importance between accessibility and availability. The thinking is that information is still in memory and forgotten but not destroyed. The information may be inaccessible given the current cue but all that is stored in memory may be potentially available with the right cue. This point is particularly important in sponsorship since sponsors believe their investment is wasted if individuals do not readily recall their brand. This may not be the case. In many market contexts, what is important is that the memory is effectively cued at important junctures.

Some time ago, Tripodi, Hirons, Bednall and Sutherland (2003) established in the sponsorship area, that like in advertising, memory for a target depends upon how a question is asked—from a psychology perspective—an examination of cueing. Questioning brings up memory paths and results in easier or more difficult attainment of a memory target depending on the competition from other remembered items. In their study respondents were asked one of the following cued recall questions: (1) “When you think of [Event Z], which sponsors come to mind?”, (2) “When you think of [Brand X], what sponsorships come to mind?”, (3) “When you think of [Category Y, e.g., banks] what sponsorships come to mind?” and then an associative recognition question (4) I am going to tell you some of Brand X’s current or recent sponsorships. For each one, could you tell me whether you were aware, before today, of Brand X sponsoring that event? Findings showed that memory for the relationship, when given an event sponsorship prompt, was twice that obtained when given the brand sponsorship prompt (32% vs. 16%). Naturally, memory for the relationship was better with additional supporting information.

This colloquial questioning format is somewhat different than that utilized in classic memory paradigms where pairs of words are memorized and instructions for response are communicated and then memory is cued with one of the members of the pair. Whereas some methods in marketing tend to invite recollection and review of alternatives, memory cues in psychology are often more purposeful in their variation so to reveal different aspects of memory. In keeping with this thinking, Cornwell et al. (2006) varied cueing direction when examining the memory support afforded by an articulating statement linking the sponsor and the event. They were expecting a two-way interaction between the presence or absence of an articulating statement and the congruence between the sponsor and the event. However, they found three-way interactions with direction of cueing. For example, when a sponsor cue was used, there was a more detrimental influence of competitor in the congruent condition on recall than in the incongruent condition where recall did not differ. In contrast, when an event cue was used, there was a moderate influence of competitor on recall in the congruent condition and a more dramatic influence in the incongruent condition. This interaction and oth-

ers demonstrated the complexity of the recall process which presumably depended on the strength of the relationship between the articulated mediator and each of the components (sponsor and event; see Nelson, McKinney, Gee, & Janczura, 1998), as well as the differential similarity between the sponsor and its competitor when the sponsor was congruent with the event as opposed to when the sponsor was incongruent with the event. None of this complexity would have been revealed if the direction of cueing had not been purposely manipulated.

As we previously noted, when preexisting relationships are involved it is very difficult to tell whether effects are being produced by encoding processes or by retrieval processes. Indeed it seems probable that preexisting knowledge, whether conceptualized as a schema or a script (e.g., what do you normally do when attending a concert or sport event), can influence both encoding and retrieval. That is, the new knowledge encountered in a situation can be linked to the preexisting knowledge and the preexisting knowledge can be used to provide cues at the time of retrieval. This ability to use preexisting knowledge to cue memory makes it very difficult in a field study to identify the cue that a person was using to answer a question about the sponsor(s) of an event. For example, is the person using the cue supplied by the interviewer (e.g., the event) or a memory for the stadium or a portion of the stadium where sponsor logos are displayed? The experienced fan may attempt to visualize that portion of the stadium in order to answer the question about the sponsor(s). This issue has important ramifications regarding whether the stadium, opera house, festival, or mass media viewing experience will transfer to a within-store display context or to other out of context retrievals.

MEMORY EXPERIENCES COMBINING WITH, DEFINING, OR REDEFINING OTHER MEMORY EXPERIENCES

We know very little about the interaction of sponsorship and advertising. In that much of sponsorship-linked marketing is event based, with exposure either via attending or viewing a broadcast, it builds autobiographical memories of past personal experiences and it may show integration tendencies similar to those found when advertising follows other experiences. For example, Braun, Ellis, and Loftus (2002) showed that autobiographical referencing advertisements, asking you to remember the past; can result in people believing that they had experiences they did not. After a delay, participants in their study who had experienced the autobiographical reference to childhood experience in the advertisement even reported greater confidence in a childhood experience of shaking hands with a character in Disneyland that had not been introduced to the park.

Postexperience advertising (Braun, 1999) is naturally an aspect of sponsorship-linked marketing, whether communications focus on the sponsorship or not. The possibility of “backward framing” advertisements may arise simply from the chronology of events and advertising scheduling. While there is clarity regarding an influence of advertising on memory, there is little agreement on exactly how this process transpires (Braun-LaTour, LaTour, Pickrell, and Loftus, 2004).

The reconsolidation hypothesis suggests that consolidated memories recalled by a reminder enter a vulnerable phase where they are sensitive to disruption after which a stabilization process returns memory to the former state (Nader, Schafe, & LeDoux, 2000). Others argue that the process of reconsolidation can be explained using principles such as contextual reinstatement and item-context binding (Sederberg, Gershman, Polyn, & Norman, 2011). To think systematically about the question, the task that requires explaining involves the learning of two lists. List A is learned one day and when the participants return a day later they learn List B. In the reminder group they are reminded of List A just before learning List B. This results in asymmetrical pattern of intrusions when delayed recall is then assessed. That is, those participants who are asked to recall List A make a lot of intrusions from List B. However, those participants who are asked to recall List B do not make many intrusions from List A. The reconsolidation hypothesis proposes that the reminder puts the List A memories into a vulnerable state where they can be modified by List B learning (Hupbach, Gomez, Hardt, & Nadel, 2007). The contextual reinstatement hypothesis proposes that the reminder reinstates the List A context which then becomes associated with the List B items. Importantly, the List B context does not become associated with the List A items producing the asymmetric pattern of intrusions. Regardless of which of these explanations is correct the important point here is that the List B experience is modifying what is retrieved when participants try and recall the List A experience.

This returns to the eyewitness literature mentioned previously. Not only is memory for an event subject to influence by advertising, it is also molded by the process of recall. In examining recall, Chan, Thomas, and Bulevich (2009) show that immediate cued recall about an event may result in a tendency to incorporate misinformation. As one could imagine, there are myriad ways that memory might be influenced, altered, enhanced, and perhaps associated attitudes would also change. At this point we have a limited understanding of how a memory interacts with an earlier memory perhaps altering or redefining the earlier remembered experience. We do, however, have some ideas about retaining and strengthening memories that would apply to indirect marketing.

RETAINING AND ENHANCING MEMORIES

As Yet, Unexamined Effects of Test-Like Experiences

Roediger and Karpicke (2006) show that simply testing someone regarding their exposure strongly supports memory. This is termed a testing effect. For example if, following an event, people logged onto a Web site where they had to answer questions before being considered for a prize would be a likely way to enhance some of the links that might have only been weakly established by the presence of the brand in the context of the event. Further, repeated testing with feedback has been shown to support academic learning better than repeated, spaced study (Larsen, Butler, & Roediger, 2009). As brands expand their cause-related sponsorships and seek to develop a valid and observable social responsibility reputation (Lee & Cornwell, 2011), the use of online, test-like reinforcement of their cause sponsorship involvement or outcomes could be valuable and affordable.

Although many companies seek to leverage social media in relationship to sponsorship, measurement of these efforts typically stops with counting the number of Facebook, Twitter, or Blog commentaries. As has been shown, online activation by the brand, especially when noncommercial in nature and related to the sponsorship yields improvements in brand attitude (Weeks, Cornwell, & Drennan, 2008), but more may be done to support memory by adding easily administered, test-like challenges.

Self-Supplied Information and Generation Effects

Another approach to support memory that lends itself to the nature of sponsorship-linked marketing is the generation effect (Slamecka & Graf, 1978). For example, generating “cold” from the antonym cue “hot-c...” results in better memory than simply reading “hot-cold.” In advertising, internally generated information (supplying a brand name to a question) has been found to be more memorable than externally presented information (e.g., hearing about a brand; Reardon & Moore, 1996). Furthermore, Reardon, Durso, Foley and McGahan (1987) showed that the better memory for generations made by those with expertise in a field since they have a more elaborate knowledge structures. This thinking is readily applied to sponsorship since many sports, events, and activities naturally self-select individuals with expertise.

Improved memory for a sponsor might be built into on-site or online activation by presenting opportunities for individuals to generate information. Finally,

exploration of generation effects possible in collateral sponsorship materials seems important because the demands of generating a response tend to focus attention on item information over contextual information (Jurica & Shimamura, 1999). This would be particularly important if the sponsor wanted to enhance sponsor recall but was less concerned about associations to the event context.

The Influence of Emotion

Within the context of sponsored sport, Bal, Quester, and Plewa (2009) make an interesting distinction between sports-related emotion and nonsports-related emotion. Their central thesis is that while the drama of an intensive sport event may detract from memory for a brand, the positive emotions stemming from the overall event experience may work differently. Emotions stemming from sponsorships may vary depending on what is sponsored (Christensen, 2006) but the extent to which emotion supports memory has not been developed in research on sponsorship. One of the distinct characteristics that sponsored events and activities are heralded as providing is an emotional connection to the brand. From a memory perspective, emotion captured, as arousal does not just create the impression of a vivid memory, it can actually support memory for context detail (Schmidt, Pooja, & Kensinger, 2011) and influence attitudes (Bee & Madrigal, 2012). As before, utilization of this strategy would depend on the strategic goals of the brand, and those utilizing event-based emotions may seek to make a strong connection to context details.

SUMMARY AND CONCLUSION

We have seen that the various measurements of memory for sponsorship relationships place different requirements on memory systems. One can see that the nature of sponsorship-linked marketing measurement has become more sophisticated and is at a dynamic junction with both academic and practitioner research seeking more complete and explanatory models of sponsorship effects.

Several remaining areas are in need of research. We still know very little about the relationship between implicit learning and behavior change. Can repeated exposure to a brand name in a sporting event context result in favorable in-store behaviors such as brand preference? If that same brand name had been paired with a music event would changes to behavior be similar?

We also know very little about the relationship between attitude and memory. Is it the case that some sponsorship-linked marketing programs establish good memories but at the same time result in negative attitudes that may counter any wished for brand-related behaviors. For example, perceptions of commercialization perhaps stemming from excessive brand logo placement and negative behaviors by those attending or

those sponsored could result in a disinclination to purchase. On the other hand, do we at times measure attitudes and believe them to have more importance than they do. For example, researchers have found negative attitudes toward ambushing but we do not know if these translate to brand-related behaviors against ambusher brands. There is a social bias against transgressors during explicit questioning regarding ambushing that may not be followed through in terms of "punishing" a brand for the transgression, especially if it were clever or entertaining.

Research on sponsorship-linked marketing communications is at a critical juncture. Research on memory is dynamic in the sense that many perspectives on what constitutes a memory goal for sponsorship and how one measures it are in revision. A search for insight is underway. The juncture is significant point in time because prior ways of conducting research in this area are being abandoned. Long-term, benchmarked models of memory, involving, not a single sponsorship, but a portfolio of sponsorships and the true complexity of these communication contexts is the path forward.

REFERENCES

- Appleton-Knapp, S. L., & Bjork, R. A. (2005). Examining the spacing effect in advertising: Encoding variability, retrieval processes, and their interaction. *Journal of Consumer Research*, 32, 266–276.
- Bal, C., Quester, P., & Plewa, C. (2009). Event-related emotions: A key metric to assess sponsorship effectiveness. *Journal of Sponsorship*, 2, 367–378.
- Balasubramanian, S. K., Karrh, J. A., & Patwardhan, H. (2006). Audience response to product placements: An integrative framework and future research agenda. *Journal of Advertising*, 35, 115–141.
- Barnes, J. M., & Underwood, B. J. (1959). Fate of first-list associations in transfer theory. *Journal of Experimental Psychology*, 58, 97–105. doi: 10.1037/h0047507
- Becker-Olsen, K., & Simmons, C. J. (2002). When do social sponsorships enhance or dilute equity? Fit, message source, and the persistence of effects. *Advances in Consumer Research*, 29, 287–289.
- Becker-Olsen, K. L., & Hill, R. P. (2006). The impact of sponsor fit on brand equity: The case of non-profit service providers. *Journal of Service Research*, 9, 73–83.
- Bee, C., & Madrigal, R. (2012). It's not whether you win or lose; it's how the game is played: The influence of suspenseful sports programming on advertising. *Journal of Advertising*, 41, 47–58.
- Braun, K. A. (1999). Postexperience advertising effects on consumer memory. *Journal of Consumer Research*, 25, 319–334.
- Braun, K. A., Ellis, R., & Loftus, E. F. (2002). Make my memory: How advertising can change our memories of the past. *Psychology and Marketing*, 19, 1–23.
- Braun-LaTour, K. A., LaTour, M. S., Pickrell, J. E., & Loftus, E. F. (2004). How and when advertising can influence memory for consumer experience. *Journal of Advertising*, 33, 7–25.

- Brown, A. S. (1976). Spontaneous recovery in human learning. *Psychological Bulletin*, 83, 321–328.
- Burke, R. R., & Srull, T. K. (1988). Competitive interference and consumer memory for advertising. *Journal of Consumer Research*, 15, 55–68.
- Burton, N., & Chadwick, S. (2009). Ambush marketing in sport: An analysis of sponsorship protection means and counter-ambush measures. *Journal of Sponsorship*, 2, 303–316.
- Chan, J. C. K., Thomas, A. K., & Bulevich, J. B. (2009). Recalling a witnessed event increases eyewitness suggestibility: The reversed testing effect. *Psychological Science*, 20, 66–73.
- Christensen, S. R. (2006). Measuring consumer reactions to sponsoring partnerships based upon emotional and attitudinal responses. *International Journal of Market Research*, 48, 61–80.
- Cornwell, T. B. (1997). The use of sponsorship-linked marketing by tobacco firms: International public policy issues. *Journal of Consumer Affairs*, 31, 238–254.
- Cornwell, T. B. (2008). State of art and science in sponsorship-linked marketing. *Journal of Advertising*, 37, 41–55.
- Cornwell, T. B., & Kwon, Y. (2013). The ascendancy of corporate sponsorship. Working Paper, University of Oregon.
- Cornwell, T. B., Roy, D. P., & Steinard, E. A. (2001). Exploring manager's perceptions of the impact of sponsorship on brand equity. *Journal of Advertising*, 30, 41–51.
- Cornwell, T. B., Weeks, C. S., & Roy, D. P. (2005). Sponsorship-linked marketing: Opening the black box. *Journal of Advertising*, 34, 21–42.
- Cornwell, T. B., Humphreys, M. S., Maguire, A. M., Weeks, C. S., & Tellegen, C. L. (2006). Sponsorship linked marketing: The role of articulation in memory. *Journal of Consumer Research*, 33, 312–321.
- Cornwell, T. B., Humphreys, M. S., Quinn, E. A., & McAlister, A. R. (2012). Memory of sponsorship-linked marketing communications: The effect of competitor mentions. *SAGE Open*, October–December 1–14, DOI: 10.1177/2158244012468139.
- Craik, F. I. M., & Lockhart, R. S. (1972). Levels of processing: A framework for memory research. *Journal of Verbal Learning and Verbal Behavior*, 11, 671–684.
- Dalakas, V., Madrigal, R., & Burton, R. (2008). Understanding ambush marketing: Implications from information processing. In L. R. Kahle & C. Riley (Eds.), *Sports marketing and the psychology of marketing communication* (pp. 293–306). Mahwah: Lawrence Erlbaum and Associates.
- Danylchuk, K. E., & MacIntosh, E. (2009). Food and non-alcoholic beverage sponsorship of sporting events: The link to the obesity issue. *Sport Marketing Quarterly*, 69–80.
- Dardis, F. E. (2009). Attenuating the negative effects of perceived incongruence in sponsorship: How message repetition can enhance evaluations of an “incongruent” sponsor. *Journal of Promotion Management*, 15, 36–56.
- Duke, C. R., & Carlson, L. (1994). Applying implicit memory measures: Word fragment completion in advertising tests. *Current Issues and Research in Advertising*, 16, 29–39.
- Dyne, A. M., Humphreys, M. S., Bain, J. D., & Pike, R. (1990). Associative interference effects in recognition and recall. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 16, 813–824.
- Einstein, G. O., & Hunt, R. R. (1980). Levels of processing and organization: Additive effects of individual-item and relational processing. *Journal of Experimental Psychology: Human Learning and Memory*, 6, 588–598.
- Friestad, M., & Wright, P. (1994). The persuasion knowledge model: How people cope with persuasion attempts. *Journal of Consumer Research*, 21, 1–31.
- Glanzer, M., & Bowles, N. (1976). Analysis of the word-frequency effect in recognition memory. *Journal of Experimental Psychology: Human Learning and Memory*, 2, 21–31.
- Hawkins, S. A., & Hoch, S. J. (1992). Low-involvement learning: Memory without evaluation. *Journal of Consumer Research*, 19, 212–225.
- Hawkins, S. A., Hoch, S. J., & Meyers-Levy, J. (2001). Low-involvement learning: Repetition and coherence in familiarity and belief. *Journal of Consumer Psychology*, 11, 1–11.
- Heathcote, A. (2006). Examining the origins of the word frequency effect in episodic recognition memory and its relationship to the word frequency effect in lexical memory. *Memory and Cognition*, 34, 826–838.
- Herrmann, J.-L., Walliser, B., & Kacha, M. (2011). Consumer consideration of sponsor brands they do not remember: Taking a wider look at the memorization effects of sponsorship. *International Journal of Advertising*, 30, 259–281.
- Hintzman, D. L. (1988). Judgments of frequency and recognition memory in a multiple-trace memory model. *Psychological Review*, 95, 528–51.
- Humphreys, M. S., Bain, J. D., & Pike, R. (1989). Different ways to cue a coherent memory system—A theory for episodic, semantic, and procedural tasks. *Psychological Review*, 96, 208–233.
- Humphreys, M. S., Maguire, A. M., McFarlane, K. A., Burt, J. S., Bolland, S. W., Murray, K. L., et al. (2010a). Using maintenance rehearsal to explore recognition memory. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 36, 147–159.
- Humphreys, M. S., Tangen, J. M., Cornwell, T. B., Quinn, E. A., & Murray, K. L. (2010b). Unintended effects of memory on decision making: A breakdown in access control. *Journal of Memory and Language*, 63, 400–415.
- Humphreys, M. S., Cornwell, T. B., McAlister, A. R., Kelly, S. J., Quinn, E. A., & Murray, K. L. (2010c). Sponsorship, ambushing, and counter-strategy: Effects upon memory for sponsor and event. *Journal of Experimental Psychology: Applied*, 16, 96–108.
- Hunt, R. R. (2013). Precision in memory through distinctive processing. *Current Directions in Psychological Science*, 22, 10–15.
- Hunt, R. R., & Einstein, G. O. (1981). Relational and item-specific information in memory. *Journal of Verbal Learning and Verbal Behavior*, 20, 497–514.
- Hupbach, A., Gomez, R., Hardt, O., & Nadel, L. (2007). Reconsolidation of episodic memories: A subtle reminder triggers integration of new information. *Learning & Memory*, 14, 47–53.
- Janiszewski, C., Noel, H., & Sawyer, A. G. (2003). A meta-analysis of the spacing effect in verbal learning: Implications for research on advertising repetition and consumer memory. *Journal of Consumer Research*, 30, 138–149.
- Jiang, N., & Ling, P. M. (2011). Reinforcement of smoking and drinking: Tobacco marketing strategies linked with alcohol in the United States. *American Journal of Public Health*, 101, 1942–1954.
- Johar, G. V., & Pham, M. T. (1999). Relatedness, prominence, and constructive sponsor identification. *Journal of Marketing Research*, 36, 123–143.

- Jones, S. C., Phillipson, L., & Barrie, L. (2009). Most men drink . . . especially like when they play sports—alcohol advertising during sporting broadcasts and the potential impact on child audiences. *Journal of Public Affairs*, 10, 59–73.
- Jurica, P. J., & Shimamura, A. P. (1999). Monitoring item and source information: Evidence for a negative generation effect in source memory. *Memory & Cognition*, 27, 648–656.
- Keller, K. L. (1993). Conceptualizing, measuring, and managing customer-based brand equity. *Journal of Marketing*, 57, 1–22.
- Keller, K. L. (2003). Brand synthesis: The multidimensionality of brand knowledge. *Journal of Consumer Research*, 29, 595–600.
- Kelly, S. J., Cornwell, T. B., Coote, L. V., & McAlister, A. R. (2012). Event related advertising and the special case of sponsorship-linked advertising. *International Journal of Advertising*, 31, 15–37.
- Krafka, C., & Penrod, S. (1985). Reinstatement of context in a field experiment on eyewitness identification. *Journal of Personality and Social Psychology*, 49, 58–69.
- Lamont, M., Hing, N., & Gainsbury, S. (2011). Gambling on sport sponsorship: A conceptual framework for research and regulatory review. *Sport Management Review*, 14, 246–257.
- Larsen, D. P., Butler, A. C., & Roediger, H. L. (2009). Repeated testing improves long-term retention relative to repeated study: a randomized controlled trial. *Medical Education*, 43, 1174–1181.
- Lee, S. P., & Cornwell, T. B. (2011). A framework for measuring the contributions of sport to society. In L. R. Kahle & A. G. Close (Eds.), *Consumer behavior knowledge for effective sports and event marketing* (pp. 31–57). New York: Taylor & Francis Group.
- Maclean, A., & Bonington, J. (2008). Sports sponsorship in the UK: The impact of regulatory intervention. *Journal of Sponsorship*, 1, 380–387.
- McAlister, A. R., Kelly, S. J., Humphreys, M. S., & Cornwell, T. B. (2012). Change in a sponsorship alliance and the communication implications of spontaneous recovery. *Journal of Advertising*, 41, 5–16.
- McDaniel, S. R., & Heald, G. R. (2000). Young consumers' responses to event sponsorship advertisements of unhealthy products: Implications of schema-triggered affect theory. *Sport Management Review*, 3, 163–184.
- McKelvey, S., & Grady, J. (2008). Sponsorship program protection strategies for special sports events: Are event organizers outmaneuvering ambush marketers. *Journal of Sport Management*, 22, 550–586.
- Meenaghan, T. (1996). Ambush marketing: A threat to corporate sponsorship? *Sloan Management Review*, 38, 103–113.
- Meenaghan, T. (2011). Mind the gap in sponsorship measurement. *ADMAP*, 524, 34–36.
- Nader, K., Schafe, G. E., & Le Doux, J. E. (2000). Fear memories require protein synthesis in the amygdala for reconsolidation after retrieval. *Nature*, 406, 722–726.
- Nelson, D. L., McKinney, V. M., Gee, N. R., & Janczura, G. A. (1998). Interpreting the influence of implicitly activated memories on recall and recognition. *Psychological Review*, 105, 299–324.
- Nelson, T. (1977). Repetition and depth of processing. *Journal of Verbal Learning and Verbal Behavior*, 16, 151–171.
- Olson, E. L., & Thjømøe, H. M. (2011). Explaining and articulating the fit construct in sponsorship. *Journal of Advertising*, 40, 57–70.
- Pitt, L., Parent, M., Berthon, P., & Steyn, P. G. (2010). Event sponsorship and ambush marketing: Lessons from the Beijing Olympics. *Business Horizons*, 53, 281–290.
- Pham, M. T., & Vanhuele, M. (1997). Analyzing the memory impact of advertising fragments. *Marketing Letters*, 8, 407–417.
- Pracejus, J. W., & Olson, G. D. (2004). The role of brand/cause fit in the effectiveness of cause-related marketing campaigns. *Journal of Business Research*, 57, 635–640.
- Quester, P. G. (1997). Awareness as a measure of sponsorship effectiveness: The Adelaide Formula One Grand Prix and evidence of incidental ambush effects. *Journal of Marketing Communications*, 3, 1–20.
- Reardon, R., & Moore, D. J. (1996). The greater memorability of self-generated versus externally presented product information. *Psychology & Marketing*, 13, 305–320.
- Reardon, R., Durso, F. T., Foley, M. A., & McGahan, J. R. (1987). Expertise and the generation effect. *Social Cognition*, 5, 336–348.
- Reber, R., Winkielman, P., & Schwarz, N. (1998). Effects of perceptual fluency on affective judgements. *Psychological Science*, 9, 45–48.
- Rifon, N. J., Choi, S. M., Trimble, C. S., & Li, H. (2004). Congruence effects in sponsorship: The mediating role of sponsor credibility and consumer attribution of sponsor motive. *Journal of Advertising*, 33, 29–42.
- Roediger, H. L. (1990). Implicit memory: Retention without remembering. *American Psychologist*, 45, 1043–1056.
- Roediger, H. L., & Karpicke, J. D. (2006). Test-enhanced learning: Taking memory tests improves long-term retention. *Psychological Science*, 17, 249–255.
- Sandler, D. M., & Shani, D. (1989). Olympic sponsorship vs. ambush marketing: Who gets the gold? *Journal of Advertising Research*, 29, 9–14.
- Schacter, D. L. (1987). Implicit memory: History and current status. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 13, 501–518.
- Schmidt, K., Pooja, P., & Kensinger, E. A. (2011). Emotion's influence on memory for spatial and temporal context. *Cognition & Emotion*, 25, 229–243.
- Schneider, L.-P., & Cornwell, T. B. (2005). Cashing in on crashing via brand placement in computer games: The effects of experience and flow on memory. *International Journal of Advertising*, 24, 321–343.
- Schumann, D. W., Petty, R. E., & Clemons, D. S. (1990). Predicting the effectiveness of different strategies of advertising variation: A test of the repetition-variation hypotheses. *Journal of Consumer Research*, 17, 192–202.
- Sederberg, P. B., Gershman, S. J., Polyn, S. M., & Norman, K. A. (2011). Human memory reconsolidation can be explained using the temporal context model. *Psychonomic Bulletin & Review*, 18, 455–468.
- Slamecka, N. J., & Graf, P. (1978). The generation effect: Delineation of a phenomenon. *Journal of Experimental Psychology: Human Learning and Memory*, 4, 592–604.
- Smith, S. M., & Moynan, S. C. (2008). Forgetting and recovering the unforgettable. *Psychological Science*, 19, 462–468.
- Tripodi, J. A., Hirons, M., Bednal, D., & Sutherland, M. (2003). Cognitive evaluation: Prompts used to measure sponsorship awareness. *International Journal of Market Research*, 45, 435–455.
- Tulving, E. (1981). Similarity relations in recognition. *Journal of Verbal Learning and Verbal Behavior*, 20, 479–496.
- Tulving, E., & Pearlstone, Z. (1966). Availability versus accessibility of information in memory for words. *Journal of Verbal Learning and Verbal Behavior*, 5, 381–391.

- Underwood, B. J. (1948). "Spontaneous recovery" of verbal associations. *Journal of Experimental Psychology*, 38, 429–439.
- Weeks, C. S., Cornwell, T. B., & Humphreys, M. (2006). Conceptualizing sponsorship: An item and relational information account. In L. R. Kahle & C.-H. Kim (Eds.), *Creating images and the psychology of marketing communications* (pp. 257–276). Mahwah, NJ: Lawrence Erlbaum and Associates.
- Weeks, C. S., Cornwell, T. B., & Drennan, J. C. (2008). Leveraging sponsorships on the internet: Activation, congruence, and articulation. *Psychology & Marketing*, 25, 637–654.
- Wheeler, M. A. (1995). Improvement in recall over time without repeated testing: Spontaneous recovery revisited. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 21, 173–184.
- Wong, C. K., & Read, J. D. (2011). Positive and negative effects of physical context reinstatement on eyewitness recall and identification. *Applied Cognitive Psychology*, 25, 2–11.
- Zajonc, R. B. (1968). Attitudinal effects of mere exposure. *Journal of Personality and Social Psychology Monograph Supplement*, 9, 1–27.
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- Correspondence regarding this article should be sent to: T. Bettina Cornwell, Lundquist College of Business 1208 University of Oregon, University of Oregon, Eugene, Oregon 97403 (tbc@uoregon.edu).